

Medikart — Product Requirements Document

Production Architecture, Business Specifications & Deployment Plan (v2.0)

Target Domains: medikart.pk | www.medikart.pk | admin.medikart.pk
Classification: Licensed Digital Healthcare Platform & Multi-Vendor E-Pharmacy
Status: Production Ready | **Published:** September 2026

1. Executive Summary & Vision

Medikart is Pakistan's premier licensed digital healthcare and e-pharmacy platform. It establishes a seamless bridge between registered physical retail pharmacies, licensed pharmacists, and retail patients nationwide. The platform provides hyper-fast intra-city delivery (2–4 hours in Lahore, Karachi, Islamabad), nationwide courier fulfillment (24–48 hours), automated 30-day chronic prescription refills, strict DRAP-compliant drug safety verification, and an AI-assisted pharmacist consultant.

Core Objectives:

- **100% Authentic Medications:** Direct integration with verified pharmacy partners with cold-chain monitoring.
- **DRAP & Narcotics Safety:** Strict demarcation between Over-The-Counter (OTC) items and Schedule X controlled drugs.
- **Dvago-Style Instant Search:** Zero-latency 3-tier search overlay categorized by keywords, products, and categories.
- **Chronic Disease Support:** Automated 30-day recurring refill scheduling with multi-channel alerts.
- **Mobile & PWA Ready:** Full Progressive Web App (PWA) with standalone home-screen installation on Android and iOS.

2. System Architecture & Tech Matrix

Medikart is deployed on a unified Hostinger VPS environment fronted by Cloudflare's global edge network.

Layer	Technology	Deployment & Role
Storefront (Web)	Next.js 14 (App Router)	Port 3000 (PM2 fork mode) — High performance SSR/SSG consumer frontend & PWA.
Backend API	Node.js 20 / Express	Port 5000 (PM2 cluster mode) — REST API, Zod validations, JWT/2FA, cron jobs.
Admin Panel	React 18 + Vite 6	Static SPA (dist/) served directly by Nginx at admin.medikart.pk.
Database	MongoDB Atlas (M0/Prod)	Managed cloud replica-set with connection pooling (10-100) and TTL cleanup.
Cache & Limits	Redis (ioredis)	Port 6379 — Cache-Aside product catalog & atomic sliding-window rate limiters.
AI Services	Google Gemini & Groq	Clinical blog generator, structured content writer & 'Medi' OTC symptom chatbot.
Edge / Proxy	Cloudflare + Nginx	Full (strict) SSL, DDoS protection, Gzip compression, Real IP restoration.

3. User Roles & Permission Model

User Role	Authentication	Permitted Scope & Capabilities
Guest Visitor	Anonymous / UUID Cookie	Browse catalog, 3-tier search, persistent cart (30-day TTL), instant order upload.
Registered Customer	JWT + Email OTP	Order history, saved addresses, wishlist, monthly refill recurring subscriptions.
Pharmacy Staff	JWT (Scoped Role)	Branch inventory management, order dispatch, branch commission ledger.
Admin / Pharmacist	JWT + RBAC	Prescription verification drawer, pricing instant orders, catalog CRUD.
Super Admin	JWT + TOTP 2FA	System governance, pharmacy onboarding, commission auditing, user management.

4. Core Functional Modules & User Stories

4.1. Dvago-Style 3-Tier Live Search & Background Blur

Provides an interactive search experience replicating top pharmacy standards in Pakistan. When the user focuses the search bar, the background blurs smoothly. The engine returns results grouped in 3 tiers: (1) *Suggested Keywords* (pre-seeded with popular authentic medicines), (2) *Matching Products* (with unit dosage, price in PKR, prescription requirement badge, and quick Add-to-Cart/Wishlist buttons), and (3) *Matching Categories* (direct links to filtered catalogs).

4.2. Prescription Verification & DRAP Narcotics Safety

Strict compliance with Pakistani drug laws and DRAP guidelines. Over-The-Counter (OTC) items can be purchased immediately. Restricted medicines require prescription upload. Controlled substances (Schedule X / Narcotics) are hard-gated: they are blocked from standard online checkout and digital card payments, requiring manual pharmacist review and COD fulfillment. Through *Instant Order* (/instant-order), customers upload a doctor slip, and a pharmacist prices the items in the admin portal.

4.3. Monthly Chronic Care Refill Program (/refill)

Engineered for chronic therapy maintenance (diabetes, cardiac care, hypertension). Patients save their recurring medicine lists. A daily background cron job (Node-cron) calculates refill dates, sending automated reminders via Mailjet 3 days prior to expiration with 1-click cart replenishment.

4.4. Multi-Vendor Pharmacy Network & Commission Ledger

Connects physical retail branches across major cities. Admin dashboard tracks per-order platform commissions vs pharmacy payouts, with an automated resilient background queue syncing order details to Google Sheets for fulfillment teams.

4.5. Clinical Content & AI Blog Engine (/blogs)

60 structured clinical medical guides with unique 1200x630 WebP photography, DRAP dosage tables, and FAQ accordions. Powered by Google Gemini 1.5 API with physician editorial oversight.

4.6. AI Pharmacist Assistant ('Medi' Chatbot)

Natural language assistant for OTC symptom checking, store navigation, and delivery FAQs. Enforces pediatric liquid formulation safety rules and an in-memory narcotic keyword redact filter.

4.7. Progressive Web App (PWA)

Fully standalone installable application with manifest.webmanifest and sw.js service worker. Features dedicated install triggers in header, mobile navigation drawer, and footer.

5. Payment Gateways & Banking Integrations

- **Cash on Delivery (COD):** Available nationwide across all cities.
- **Kuickpay Hosted Gateway:** PCI-DSS SAQ-A compliant credit/debit card processing with HMAC-SHA256 webhook validation.
- **1Bill / Kuickpay & Meezan Bank:** Seamless bill payment via consumer invoice ID directly from the Meezan Bank mobile app.

6. Non-Functional Requirements (NFRs)

Category	Standard & Specification
Security	JWT 32+ char secrets, bcrypt password hashing, TOTP 2FA, Helmet security headers, Nginx rate limits.
File Safety	Magic-byte header inspection (Sharp for images, %PDF- for documents); 5MB/10MB limits; public directory browsing blocked for prescriptions.
Performance	Redis Cache-Aside, Next.js SSR parallelization, Gzip compression, Nginx proxy buffering.
Concurrency	PM2 cluster mode scaling across CPU cores with primary-worker-only cron registration to prevent multi-worker cron collisions.
Data Privacy	Deep recursive log sanitization scrubbing JWTs, passwords, card CVVs, and OTPs from production server logs.

7. Production Infrastructure & Deployment

- Domain:** medikart.pk | **Server:** Hostinger VPS (Ubuntu/Debian) | **DNS:** Cloudflare (Proxied )
- **Certbot Order:** Stop Nginx → Issue standalone certs for medikart.pk, www, admin → Enable medikart.conf → Restart Nginx.
 - **PM2 Cluster:** pm2 start nginx/ecosystem.config.js orchestrates Express API (:5000) and Next.js (:3000).

Document Authority: Medikart Technical Leadership	Version: 2.0 (Production Release)
Status: Approved for Deployment	Target Launch: Q3 2026